

DISTRIBUTION MATCHING IS NOT TASK PRESERVATION: AUDITING GENERATIVE MODELS INSIDE DATA PIPELINES

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ABSTRACT

Generative models increasingly sit inside data pipelines rather than at the end, aligning data from different sources or manufacturing training examples. They are picked by asking whether outputs resemble the data we want. The failure is built into that question. The audit is simple: freeze the model that consumes the data, score it alongside the resemblance score on identical inputs, and add a control with no learned prior. Passing data through a learned transform erases information the task needs while resemblance improves. The damage belongs to the objective, not an architecture: it holds for both adversarial and diffusion generators, grows with how much of the transform is applied, survives constraints meant to preserve structure, and persists even when resemblance worsens. The mechanism is selectivity: distribution matching rewards reproducing common patterns while rare ones are overwritten, and rare ones often carry the label. Without a learned prior nothing is substituted, and the task survives. Two unrelated fields, driving and clinical, behave identically: harmonizing hospital scans toward the sites a tumor segmenter was trained on drops its accuracy from 0.68 to 0.06, while plain normalization holds 0.67. The data looks better; the model has stopped working. One frozen model and one control catch this before deployment.

1 INTRODUCTION

Generative models increasingly sit *inside* data pipelines rather than at their end: they align data collected under different conditions, or manufacture training examples outright. Practitioners choose among them by asking how closely the output resembles the data they want, typically with a distributional metric such as the Fréchet Inception Distance (FID). The implicit promise is that a closer match makes the data more *useful* to whatever consumes it downstream. We test that promise directly and find it false whenever the transform is learned: the resemblance score improves while the information the downstream task depends on is destroyed. Because the metric being optimised is precisely the one that cannot see the damage, the failure is silent.

Our contributions are:

- A **controlled protocol** for separating fidelity from utility: a frozen downstream evaluator plus a matched non-learned control, measuring FID and the task metric on identical inputs (Section 3).
- Evidence that the fidelity–utility dissociation is a property of the **learned distribution-matching objective**, not a single architecture—it reproduces under both a GAN (CycleGAN) and a diffusion model (SDEdit).
- A **continuous characterization** via SDEdit strength: a monotonic FID-vs-mIoU frontier with tight seed variance, an empty-prompt ablation that decouples utility loss from fidelity gain, and a structure-preserving (ControlNet) baseline that fails to escape the frontier.
- **Cross-domain** generality: the same dissociation in driving scenes and in clinical brain-tumor MRI.

2 RELATED WORK

Three separately-established observations converge in our claim; we unify them.

Fidelity metrics mis-rank generators for downstream tasks. Konz et al. (2024) show no perceptual metric reliably tracks segmentation quality in medical translation; Dutt et al. (2025) show low FID does not predict clinical utility; and Kynkäänniemi et al. (2023) show FID is gameable by aligning ImageNet-class statistics. These are *metric-evaluation* studies; we add a learned/non-learned control and a second generative family and domain.

Distribution-matching translation alters task content. Cohen et al. (2018) show CycleGAN-style matching can hallucinate or remove tumors to fit the target distribution, and semantic-consistency losses (Hoffman et al., 2018; Peng et al., 2023) exist precisely to patch this. We show that *without* such non-distributional anchoring the pure learned objective degrades utility, reframing those losses as evidence for our thesis.

Non-learned alignment rivals learned translation. Yang & Soatto (2020) reach strong segmentation adaptation with a non-learned spectral swap. We use histogram/color transfer as the analogous control and, unlike FDA, measure the fidelity axis explicitly to expose the dissociation. Richter et al. (2022) likewise treat a segmentation-stability metric as a guardrail that realism must not break.

Positioning. No prior work runs the clean 2×2 {learned, non-learned} $\times$ {fidelity $\downarrow$?, utility $\downarrow$?} as its thesis, ties the failure to the objective across both a GAN *and* a diffusion model, and does so across driving and medicine. This is a measurement/positioning contribution, not a new architecture.

3 SETUP: SEPARATING FIDELITY FROM UTILITY

Given source images x and a translation T , we evaluate $T(x)$ on two axes. **Fidelity** is $\text{FID}(T(x), \text{target})$ via clean-fid (Parmar et al., 2022); lower is closer to the target. **Utility** is a *frozen* downstream model f (never retrained) applied to $T(x)$ and scored against ground truth; because f is fixed, any change in its score is attributable purely to T . We compare four families of T : (i) identity (raw source, reference); (ii) non-learned (histogram / color transfer); (iii) learned-GAN (CycleGAN, Zhu et al., 2017); and (iv) learned-diffusion (SDEdit, Meng et al., 2022, over a sweep of noise strengths, plus ControlNet-Canny, Zhang et al., 2023). FID is compared only at equal sample size, which it is known to bias.

4 EXPERIMENTS

4.1 DRIVING: GTA5 $\rightarrow$ CITYSCAPES

The source is GTA5 (Richter et al., 2016) and the target is Cityscapes (Cordts et al., 2016). The frozen evaluator is a Cityscapes-finetuned SegFormer-b4 (Xie et al., 2021) (19 classes); utility is mIoU. We use $N=1000$ GTA5 images per condition, with the Cityscapes validation set (500 images) as the fixed FID reference, and identical evaluator and clean-fid (GPU) for all conditions. SDEdit runs on Stable Diffusion (Rombach et al., 2022) and rows report mean $\pm$ std over three seeds {42, 43, 44}. Results are in Table 1 and Figure 1.

Findings. Figure 2 shows representative translations. (a) Learned translation lowers FID relative to raw while collapsing mIoU by 15–72%; CycleGAN more than halves FID (137 $\rightarrow$ 65) at mIoU 0.167. (b) The non-learned baseline does not improve FID yet preserves mIoU (−1.8%). (c) SDEdit strength traces a **monotonic frontier** with tight seed variance ($\sigma(\text{mIoU}) \leq 0.005$): as strength rises, FID falls (145 $\rightarrow$ 88) and mIoU falls with it (0.323 $\rightarrow$ 0.106); no learned setting reaches the ideal corner, occupied only by raw/non-learned. (d) **Empty-prompt ablation.** With no text prompt, FID is *worse* than raw (154.2 $>$ 137.2) yet mIoU still collapses (0.170)—the utility loss is the diffusion prior itself, not text guidance, and occurs even absent any fidelity gain. (e) **Structure-preserving baseline.** Conditioning on the source image’s Canny edges (ControlNet-Canny) does *not* rescue utility: it lands on the learned frontier (FID 133.2, mIoU 0.221 $\pm$ 0.002), Pareto-dominated by

Table 1: GTA5→Cityscapes, $N=1000$, frozen SegFormer-b4 + clean-fid. SDEdit rows are mean $\pm$ std over three seeds. Learned translation lowers FID while collapsing mIoU; the non-learned baseline preserves mIoU without improving FID; no learned setting reaches the low-FID/high-mIoU corner.

condition	family	FID $\downarrow$	mIoU $\uparrow$	Δ mIoU
raw GTA5	baseline	137.2	0.380	—
color match	non-learned	165.9	0.363	-0.018
SDEdit 0.20	diffusion	145.1 $\pm$ 1.8	0.323 $\pm$ 0.003	-0.057
SDEdit 0.30	diffusion	133.1 $\pm$ 2.1	0.289 $\pm$ 0.002	-0.092
SDEdit 0.40	diffusion	116.4 $\pm$ 1.1	0.237 $\pm$ 0.003	-0.143
SDEdit 0.50	diffusion	100.3 $\pm$ 1.4	0.194 $\pm$ 0.005	-0.187
SDEdit 0.60	diffusion	90.8 $\pm$ 2.0	0.150 $\pm$ 0.001	-0.231
SDEdit 0.70	diffusion	88.0 $\pm$ 0.4	0.106 $\pm$ 0.001	-0.274
CycleGAN	learned-GAN	65.5	0.167	-0.214
SDEdit 0.50, empty prompt	diffusion (abl.)	154.2 $\pm$ 1.9	0.170 $\pm$ 0.003	-0.211
ControlNet-Canny	diffusion + edge	133.2 $\pm$ 0.3	0.221 $\pm$ 0.002	-0.159

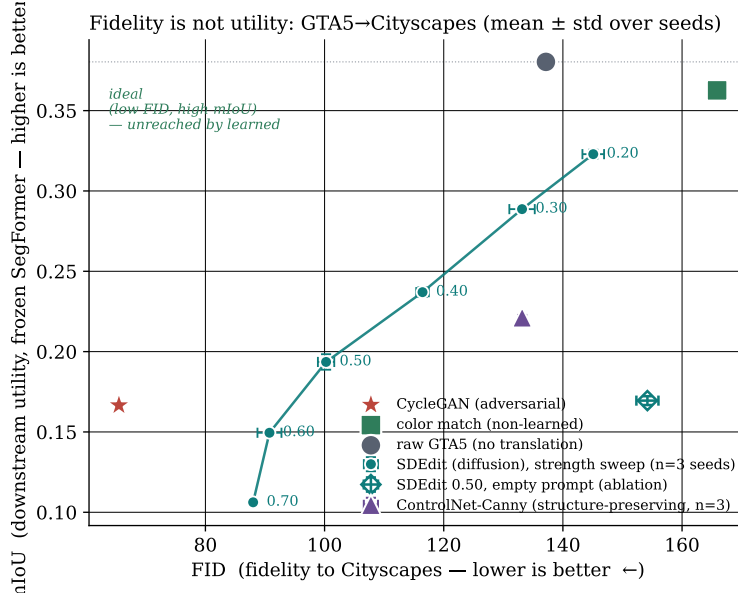


Figure 1: The fidelity–utility plane for GTA5→Cityscapes (mean $\pm$ std over seeds). The SDEdit strength sweep traces a monotonic frontier: as strength rises, FID falls and mIoU falls with it. Every learned point (CycleGAN, all SDEdit strengths, the empty-prompt ablation, and edge-conditioned ControlNet) sits well below raw and the non-learned baseline on utility; the ideal low-FID/high-mIoU corner is empty.

SDEdit at equal FID (SDEdit 0.30: FID 133.1, mIoU 0.289). Edge structure is not the task-relevant signal; only conditioning on the segmentation *map* (Peng et al., 2023) recovers utility, and only by injecting the labels—the non-distributional constraint our thesis requires. A local $N=200$ run reproduces the same ordering on an independent machine.

4.2 MEDICAL: MULTI-SITE BRAIN-TUMOR MRI

The frozen evaluator is an nnU-Net (Isensee et al., 2021) trained on BraTS (Menze et al., 2015) and frozen; utility is tumor Dice on the external UPenn-GBM site (Bakas et al., 2022) ($n=103$). The learned harmonizer is SA-CycleGAN-2.5D, a self-attention CycleGAN over 2.5D multi-modal MRI. Table 2 reports the result.



Figure 2: Qualitative GTA5→Cityscapes translations (three scenes, seed 42). Left to right: raw source, non-learned color match, CycleGAN, SDEdit at increasing strength, the empty-prompt SDEdit, and edge-conditioned ControlNet. As the learned translations become more Cityscapes-like they visibly hallucinate and alter scene structure—buildings, lane markings, added objects—that the frozen segmenter depends on, while the non-learned baseline leaves structure intact.

Table 2: External UPenn-GBM tumor Dice under a frozen BraTS-trained nnU-Net. Learned harmonization collapses Dice even though it makes images more source-like; the non-learned baseline preserves it.

input to frozen nnU-Net	family	Dice (fg. mean)
BraTS test (in-domain ref.)	—	0.771
UPenn raw (external)	baseline	0.682
UPenn + histogram match	non-learned	0.669
UPenn + learned harmonization	learned-GAN	0.064
UPenn + symmetric seg-consistency	learned-GAN (+ content)	0.205

Findings. Learned harmonization collapses external-site Dice (0.682→0.064) while the non-learned baseline preserves it (0.669). Content/segmentation-consistency losses reduce but do not eliminate the loss (0.205 $\ll$ 0.682), consistent with the driving result that the degradation is intrinsic to the learned objective. A second external site (RHUH-GBM, Cepeda et al., 2023, $n=40$) confirms the direction (0.718→0.599, −16.6%). We note two evaluation protocols exist: the headline numbers use the frozen independent nnU-Net; a separate method-development evaluator ranks the symmetric loss as the best-preserving *learned* harmonizer, which belongs to a companion method paper.

4.3 WHERE THE DAMAGE FALLS

The headline mIoU collapse hides structure. Breaking IoU out per class and grouping the 19 Cityscapes classes into large amorphous regions and thin, small, label-carrying objects (Table 3) shows the degradation is highly *selective*. The non-learned control loses a few percent on both groups, almost uniformly. Every learned translation instead loses far more on the thin classes than on the large ones: SDEdit at strength 0.50 costs 27% of large-region IoU but 83% of thin-object IoU, and CycleGAN 50% against 70%. Poles, traffic lights, traffic signs, riders and motorcycles are precisely the structures a segmenter needs and precisely what the learned prior does not preserve. Edge-conditioned ControlNet sits in between (27% / 63%): conditioning on edges recovers some thin structure, consistent with Section 3’s argument that what is missing is a non-distributional constraint, but not enough to escape the frontier.

Table 3: Per-class IoU grouped into large amorphous regions (road, sidewalk, building, vegetation, terrain, sky) and thin/small label-carrying objects (pole, traffic light, traffic sign, rider, motorcycle); classes whose raw IoU is below 0.05 are excluded, since a ratio off a near-zero baseline is noise. The non-learned control degrades both groups mildly and almost uniformly; every learned translation degrades the thin classes far more than the large ones.

group	raw	color match	CycleGAN	SDEdit 0.50	ControlNet
large regions	0.661	0.630 (−5%)	0.328 (−50%)	0.481 (−27%)	0.482 (−27%)
thin / small objects	0.174	0.168 (−4%)	0.052 (−70%)	0.030 (−83%)	0.065 (−63%)

5 ANALYSIS

The learned objective optimizes appearance under a distribution/adversarial or denoising target with no term guaranteeing task-relevant signal survives. SDEdit strength is precisely the amount of learned prior injected, and utility falls monotonically with it. The empty-prompt result decouples utility loss from fidelity gain: passing through the prior damages task content even when the output is *less* target-like. Non-learned transforms, lacking a learned prior, cannot introduce this failure; edge-conditioned ControlNet, injecting structure but not semantics, does not avoid it.

6 LIMITATIONS

FID is sample-size biased; all conditions use the same N (1000 translated vs the 500-image Cityscapes validation set as the fixed reference), and SDEdit points report mean $\pm$ std over three seeds ($\sigma(\text{mIoU}) \leq 0.005$, $\sigma(\text{FID}) \leq 2.1$), so the ordering is robust to both sample size and seed. The structure-preserving baseline uses edge (Canny) conditioning; a segmentation-map-conditioned variant would recover utility only by injecting the labels, which concedes the thesis rather than refuting it.

7 CONCLUSION

Fidelity is not utility. Rewarding distributional realism can actively harm the downstream task for learned translation, across architectures and domains. We recommend reporting a frozen-evaluator utility axis alongside any fidelity metric, and treating non-learned baselines as first-class controls.

REPRODUCIBILITY STATEMENT

All conditions regenerate from a single command; seeds are fixed and logged; frozen evaluators are referenced by checkpoint id (SegFormer-b4, nnU-Net); FID uses clean-fid at a fixed N . Code, configs, and the $N=1000$ result set are released with the paper.

A FULL PER-CLASS IOU

Table 4 reports every Cityscapes class, the breakdown behind Table 3. SDEdit and ControlNet columns are means over the three seeds. Classes with very small raw IoU (*train*, *bicycle*) are near-absent in the GTA5 subset; a relative change computed off such a baseline is noise, which is why the grouped statistics in Table 3 exclude classes with raw IoU below 0.05. The selectivity is visible directly: *road* loses little under every condition, while *traffic light*, *traffic sign*, *person* and *rider* are almost entirely destroyed by the learned translations and almost untouched by the non-learned control.

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Table 4: Per-class IoU under a frozen SegFormer-b4, $N=1000$.

class	raw	color match	CycleGAN	SDEdit 0.50	ControlNet
road	0.937	0.932	0.843	0.845	0.884
sidewalk	0.540	0.502	0.202	0.280	0.309
building	0.711	0.681	0.356	0.504	0.505
wall	0.356	0.315	0.089	0.077	0.094
fence	0.102	0.111	0.020	0.021	0.031
pole	0.269	0.252	0.112	0.095	0.159
traffic light	0.073	0.067	0.008	0.008	0.012
traffic sign	0.069	0.063	0.012	0.017	0.022
vegetation	0.649	0.625	0.364	0.455	0.427
terrain	0.227	0.159	0.083	0.050	0.011
sky	0.901	0.883	0.121	0.750	0.755
person	0.270	0.240	0.035	0.024	0.111
rider	0.182	0.156	0.028	0.015	0.085
car	0.737	0.725	0.384	0.226	0.447
truck	0.666	0.608	0.198	0.197	0.175
bus	0.248	0.259	0.206	0.097	0.127
train	0.002	0.002	0.000	0.001	0.001
motorcycle	0.279	0.303	0.101	0.016	0.046
bicycle	0.008	0.007	0.001	0.001	0.004

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